

Authentication Flow

1. JWT Token Generation:

- **Endpoint:** /api/auth/login (POST)

- **Request Body:**

```
{  
  
  "username": "user",  
  
  "password": "password"  
}
```

- **Response:**

```
{  
  
  "token": "your-jwt-token-here"  
}
```

2. JWT Token Usage:

- Add the token in the Authorization header for each protected endpoint:

Authorization: Bearer <your-jwt-token-here>

API Endpoints

1. User-User Recommendation

- **Endpoint:** /api/recommendations/<user_id>
- **Method:** GET
- **Description:** Fetch book recommendations for a specific user based on their past preferences (User-User Collaborative Filtering).
- **Authentication Required:** Yes (JWT Token in the Authorization header)
- **URL Params:**
 - user_id: The ID of the user requesting the recommendations (integer).
- **Query Params:**

- **top_n:** The number of recommendations to return (default is 5).
- **Request Example:**

```
GET /api/recommendations/1?top_n=10
```
- **Response:**

```
{
  "user_id": 1,
  "recommendations": [
    "Book Title 1",
    "Book Title 2",
    "Book Title 3" ]
}
```
- **Description:**

The system will use User-User collaborative filtering to recommend books that users with similar preferences have liked.

2. Top K Recommendations for a User

- **Endpoint:** /api/recommendations/<user_id>/top_k
- **Method:** GET
- **Description:** Retrieve the top K book recommendations for a user based on their preferences (User-User Collaborative Filtering).
- **Authentication Required:** Yes (JWT Token in the Authorization header)
- **URL Params:**
 - **user_id:** The ID of the user requesting the recommendations (integer).
- **Query Params:**
 - **count:** The number of top recommendations to fetch (default is 10).
- **Request Example:**

GET /api/recommendations/1/top_k?count=5

- **Response:**

```
{  
  
  "user_id": 1,  
  
  "top_k_recommendations": [  
  
    "Book Title 1",  
  
    "Book Title 2",  
  
    "Book Title 3"]  
  
}
```

- **Description:**

- This endpoint returns the most relevant books based on collaborative filtering for the given user.

3. Item-Item Recommendation

- **Endpoint:** /api/recommendations/item-item
- **Method:** GET
- **Description:** Fetch book recommendations based on a specific book title using Item-Item Collaborative Filtering.
- **Authentication Required:** Yes (JWT Token in the Authorization header)
- **Query Params:**

- title: The title of the book for which recommendations are needed (string).
- top_n: The number of recommendations to return (default is 5).

- **Request Example:**

GET /api/recommendations/item-item?title=The Great Gatsby&top_n=5

- **Response:**

```
{  
  
  "input_title": "Book Title",
```

```
    "recommendations": [  
        "Book Title 1",  
        "Book Title 2",  
        "Book Title 3"]  
    }  
}
```

- **Description:**

- Recommends books similar to the input book based on Item-Item collaborative filtering.

4. Content-Based Recommendation

- **Endpoint:** /api/recommendations/content-based
- **Method :** GET
- **Description:** Get book recommendations based on the content of a given book using content-based filtering.
- **Authentication Required:** Yes (JWT Token in the Authorization header)
- **Query Params:**
 - title: The title of the book for which content-based recommendations are needed (string).
 - top_n: The number of recommendations to return (default is 5).

- **Response:**

```
{  
  
    "input_title": "Book Title",  
  
    "recommendations": [  
        "Book Title 1",  
        "Book Title 2",  
        "Book Title 3",  
        "Book Title 4",  
        "Book Title 5"  
    ]  
}
```

"Book Title 3"]

}

- **Description:**

- Provides book recommendations based on content similarity (e.g., genre, plot keywords) with the specified book.

5. Get Book Details

- **Endpoint:** /api/book/<int:book_id>

- **Method:** GET

- **Description:** Fetch detailed information about a specific book based on its ID.

- **Authentication Required:** Yes (JWT Token in the Authorization header)

- **URL Params:**

- book_id: The ID of the book whose details you want to retrieve (integer).

- **Request Example:**

GET /api/book/1

- **Response:**

{

 "book_details": {

 "title": "Book Title",

 "author": "Author Name",

 "rating": 4.5,}

}

- **Description:**

- Provides detailed information about a specific book, including the title, author, rating.

Rate Limiting

- To ensure fair use of the API, all endpoints are rate-limited to **100 requests per hour**. If you exceed the limit, you'll receive an error message indicating you've hit the rate limit.

Error Handling

- Common errors are returned in the following format:

```
{  
  "error": "Error message"  
}
```

- Example Errors:

- **Unauthorized Access** (Invalid or missing JWT token):

```
{  
  "error": "Unauthorized access"  
}
```

- **Book Not Found** (Invalid book title or ID):

```
{  
  "error": "Book not found"  
}
```

Integration Notes

- **Frontend:**
 - For authentication, included the JWT token in the Authorization header when making requests to protected endpoints.

- The endpoints for recommendations can be called with the appropriate parameters to display recommendations based on user or item preferences.
- **Backend:**
 - Ensured that the Models directory contains the required models (user_user_cf_knn_model.pkl, item_item_cf_knn_model.pkl, etc.) and that the dataset is correctly loaded into books_cleaned.csv.